



Machine Learning Prescriptions

Researchers develop an AI that aids in identifying and repurposing existing drugs to combat COVID-19. <http://digit.in/mar21-31>



Herd Immunity Against COVID

New study estimates 1.7M vaccines are required to achieve herd immunity to COVID-19 in Travis County. <http://digit.in/mar21-32>

remdesivir is inconvenient for the patients and comes with a lot of side effects. The finding can potentially reduce the regimen to a single pill.

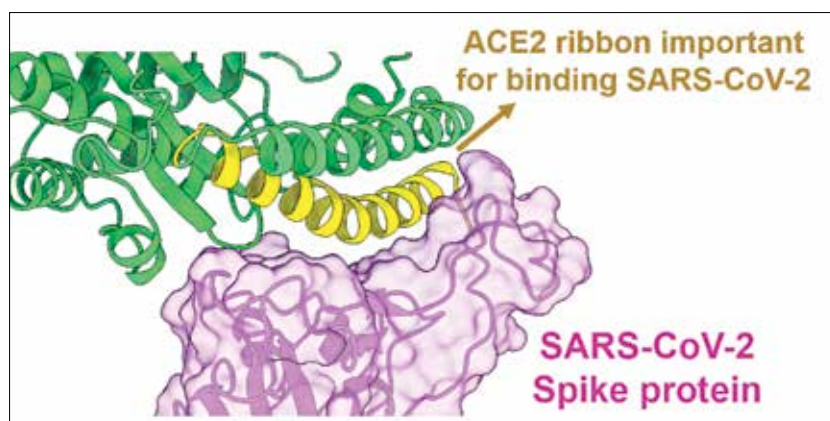
Researchers from the University of Chicago also studied how remdesivir works on a molecular level. The researchers used computer simulations to model the action of three similar drugs that are meant to inhibit the replication of the virus, remdesivir, ribavirin, and favilavir. The researchers found that while remdesivir binds strongly to the virus, ribavirin and favilavir do not. Apart from its action on the RNA polymerase, remdesivir also disrupts the protein complex in the coronavirus, further inhibiting its ability to replicate. After understanding the action of the drug on the molecular level, researchers hope to come up with better delivery techniques to make the drug more effective.

DECOY TARGET

The coronavirus accesses the cell when its spike proteins bind to a target receptor protein on the cell wall known as ACE2. This particular receptor is abundant in the cells of humans and other species, which is what makes the coronavirus so good at spreading within populations, as well as jumping species. Advancements in protein crystallisation and microscopy has now made it possible for researchers to create computer models of protein structures, and zoom in and out of their interactions. Ohio State University researchers focused on the binding location, and found some ribbon like structures that were in the tail of ACE2. These structures were the starting point for newly developed and tested peptides, or protein fragments that are similar enough to ACE2, that the spike protein binds to the peptide instead of a cell. The researchers focused on creating the peptides that were of the shortest length, but still had sufficient surface area to bind effectively with the spike protein. The team narrowed down on two

promising peptides that were effective at slowing down the coronavirus, a short peptide with minimal contact points, and a longer one. The idea of the project is to block the SARS-CoV-2 virus from binding to the cell, because it is already binded to a peptide, but in order for this method to work, the peptides have to get access to the virus before the virus gets access to the cell. The team is looking to deliver the peptides in the form of a nasal spray or a surface disinfectant. The researchers now hope to assess the capabilities of the peptides against emerging mutations of the

pandemic, bats and pangolins were two of the main suspects, and these two continue to be potential candidates for the origin of the coronavirus. Researchers have looked at the viruses existing in bat and pangolin populations, as well as the antibodies in their blood, but not yet identified a source. This is not surprising considering that the origin of SARS was not discovered for over a decade after the pandemic. What all this goes to show is that humans are not the only species that we need to be worried about, and that pets, livestock and wildlife are all at risk of zoonotic transmis-



Interaction between an ACE2 ribbon and the SARS-CoV-2 spike protein

virus, as well as move to developing the technology for therapeutic purposes. Other teams have also developed similar polymers, including one group of researchers who focused on self sterilising polymers. They are developing a “virus proof” material that can be coated on walls, countertops, railings, textiles and other high contact surfaces.

ZOONOTIC SHIFT

Dogs, cats, minks, tigers, lions and gorillas have all tested positive for infection by the SARS-CoV-2 virus. It is very likely that in these cases, the virus was transmitted to them by humans. The World Health Organisation also noted that the virus most likely jumped from an animal species to humans in the first place, although the species has not yet been identified. In the early days of the

sion, and humans may be exposed to further risk because of the tendency of the virus to hop between species. It would be hard to contain an outbreak that transmits from humans to wildlife. The species that the coronavirus has affected the most is the mink, with the Danish government culling 17 million animals in mink farms. Sequencing the DNA of virus samples collected from the mink told the story of the spread. The virus had spread from humans to minks, and then back into humans, forming a loop. Additionally, the virus acquired mutations as it spread. A vulnerable mink population is exactly the kind of thing that the virus needs to become stronger and badder. The good news is that researchers are working on vaccines for animal populations too, just as they are working on vaccines for humans. 